Grouped Permutation Feature Importance under Full-Sample In-Sample Detrending

Atmospheric Demand & Heat | Hydrological Supply | Soil Moisture Dynamics | Thermal & Radiation Regimes | Memory & Spatial Baseline

(a) ElasticNet (+Lag) [Full Trend]

- Vapor Pressure Deficit (VPD): 8.0%
- Reference Evapotranspiration (ET0): 2.5%
- Moderate Heat Days (>=30C)
- Extreme Heat Days (>=35C): 0.3%
- Extreme Degree Days (>30C): 12.7%
- Cumulative Precipitation (P): 13.3%
- Climatic Water Balance (P - ET0): 2.9%
- Topsoil Moisture (0-7 cm): 11.6%
- Mid Subsoil Moisture (7-28 cm): 5.2%
- Deep Subsoil Moisture (28-100 cm): 2.5%
- Root-Zone Soil Moisture: 1.1%
- Mean Temperature (T_mean)
- Max Temperature (T_max): 8.1%
- Min Temperature (T_min): 3.4%
- Growing Degree Days (GDD): 1.2%
- Dewpoint Temperature (T_dew): 19.5%
- Surface Solar Radiation (SSRD): 6.1%
- Lagged Yield Memory (ε_{t-1}): 0.4%
- Spatial Coordinates (Lat / Lon): 1.3%

0% 10% 20% 30%

Information Share (%)

(b) XGBoost (+Lag) [Full Trend]

- Vapor Pressure Deficit (VPD): 12.9%
- Reference Evapotranspiration (ET0): 0.9%
- Moderate Heat Days (>=30C)
- Extreme Heat Days (>=35C): 3.5%
- Extreme Degree Days (>30C): 1.6%
- Cumulative Precipitation (P): 20.9%
- Climatic Water Balance (P - ET0): 13.4%
- Topsoil Moisture (0-7 cm): 9.0%
- Mid Subsoil Moisture (7-28 cm): 5.2%
- Deep Subsoil Moisture (28-100 cm): 0.4%
- Root-Zone Soil Moisture: 2.6%
- Mean Temperature (T_mean)
- Max Temperature (T_max): 4.1%
- Min Temperature (T_min): 1.1%
- Growing Degree Days (GDD): 2.0%
- Dewpoint Temperature (T_dew): 5.7%
- Surface Solar Radiation (SSRD): 12.5%
- Lagged Yield Memory (ε_{t-1}): 4.0%
- Spatial Coordinates (Lat / Lon)

0% 10% 20% 30%

Information Share (%)

(c) LSTM (Surrogate) [Full Trend]

- Vapor Pressure Deficit (VPD): 63.5%
- Reference Evapotranspiration (ET0)
- Moderate Heat Days (>=30C)
- Extreme Heat Days (>=35C): 1.4%
- Extreme Degree Days (>30C)
- Cumulative Precipitation (P): 18.4%
- Climatic Water Balance (P - ET0)
- Topsoil Moisture (0-7 cm)
- Mid Subsoil Moisture (7-28 cm)
- Deep Subsoil Moisture (28-100 cm)
- Root-Zone Soil Moisture
- Mean Temperature (T_mean)
- Max Temperature (T_max)
- Min Temperature (T_min)
- Growing Degree Days (GDD)
- Dewpoint Temperature (T_dew): 16.7%
- Surface Solar Radiation (SSRD)
- Lagged Yield Memory (ε_{t-1})
- Spatial Coordinates (Lat / Lon)

0% 10% 20% 30%

Information Share (%)